

NEXUS

NeuroEndocrine-Gut AXes Understanding and Steering

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What if we could treat the upstream causes of women's chronic disease, addressing multiple conditions with personalized interventions?

The Opportunity

Fifty million American women suffer from conditions that medicine treats as separate diseases, while they are systemic chronic conditions: endocrine-metabolic dysfunctions such as Poly Cystic Ovarian Syndrome (PCOS) and endometriosis, bone loss, as well as cardiovascular and neurological diseases. But these are not isolated problems. They share common upstream drivers rooted in the Brain-Ovarian-Gut network, and treating those drivers can address multiple conditions simultaneously.

Three biological systems constantly communicate: the reproductive system (Hypothalamic-Pituitary-Ovarian [HPO] axis) regulating hormones like estrogen, LH, and FSH, the stress system (Hypothalamic-Pituitary-Adrenal [HPA] axis) regulating cortisol, and the gut (Microbiota-Gut-Brain [MGB] axis) including microbiome metabolites that influence both other axes. When this network becomes dysregulated, the effects cascade. Consider Follicle Stimulating Hormone (FSH): as it rises during perimenopause, it doesn't just signal reproductive aging. FSH receptors on bone cells accelerate resorption. FSH receptors on fat cells promote visceral accumulation. FSH receptors on neurons contribute to inflammation. One signal, multiple downstream effects.

Today's medicine treats each condition separately: one drug for bone loss, another for metabolic syndrome, another for menopausal symptoms. But if we can identify and correct the upstream network dysregulation, one personalized intervention could treat current symptoms while reducing risk of multiple future conditions. This is not prevention in the abstract, but treating the root cause with cascade benefits.

What NEXUS Will Build

NEXUS will create tools to decode each patient's network state and steer them to the right intervention:

- 1. Multi-Modal Sensing.** A wearable platform that continuously monitors all three systems: stress hormones (cortisol dynamics and rhythms), reproductive hormones (LH, FSH, estrogen, progesterone pulse patterns), and gut function (breath-based metabolites, validated by periodic stool and blood sampling). Critically, we capture dynamics, not just levels. LH pulse frequency and cortisol rhythms encode clinical information invisible to a single blood draw. Today, no dataset captures continuous multi-axis physiological data in women. The multi-modal data collected in NEXUS will fill that gap with an open data commons, a resource that researchers and innovators can interrogate for new targets, models, and tools for years beyond the program.
- 2. Causal Network Modeling.** Computational models that serve two complementary purposes: First, they classify patients by which system is driving their dysfunction (stress-dominant, hormone-dominant, gut-dominant, or mixed), enabling better patient stratification. Second, they mechanistically identify druggable targets by distinguishing true drivers from downstream effects, revealing opportunities for drug repurposing and novel target discovery. We call this approach "perturb and listen": every intervention reveals network structure, improving the model for the next patient. These models, trained on women-specific multi-modal data, will serve as a screening tool for testing new drug candidates against patient subtypes to predict responders before costly trials.
- 3. Personalized Interventions.** NEXUS takes a modality-agnostic approach to intervention, selecting the right treatment based on each patient's network state rather than a single drug class. This includes: drug repurposing (e.g. GLP-1s or NK3R antagonists to be repurposed for PCOS), gut-targeted therapies (microbiome therapies that modulate how gut bacteria metabolize hormones), novel targets identified by

network analysis (e.g. anti-FSH antibodies for combined bone and metabolic effects), and personalized hormone therapy that adapts dosing and timing to each woman's specific network dynamics.

Why Now

Multiple breakthroughs have converged:

- Continuous hormone sensing is validated. Researchers at Caltech, Imperial College, and startups like Level Zero Health have demonstrated wearable sensors that track reproductive hormones and cortisol in interstitial fluid with strong correlation ($r > 0.85$) to gold-standard measurements. For the first time, we can capture the pulse patterns that encode critical health information.
- Causal AI can find root causes. New machine learning methods distinguish true cause-and-effect from correlation. When we give a drug and observe how cortisol, LH, and gut inflammation respond, these models identify which network connections are driving dysfunction and which interventions will work for each patient.
- The gut-hormone connection is measurable. A January 2025 study demonstrated that breath sensors track gut microbiome activity with over 70% accuracy. Combined with validated markers of gut inflammation, we can finally bring the gut axis into continuous monitoring.

What Makes NEXUS Different

Several well-funded programs address related problems. Wellcome Leap's CARE program (\$50M) focuses on the menopause-to-cognitive-decline pathway. XPRIZE Ovarian Health (\$50M) emphasizes measuring ovarian function. ARPA-H's Delphi program builds a next-generation biosensor chiplet system.

NEXUS occupies distinct territory: We focus on metabolic and bone outcomes. We include the gut axis, which no other program addresses. We emphasize drug repurposing and therapeutic development, not just diagnostics. And we build the interpretation and treatment layer that translates sensing into personalized interventions. Most importantly, NEXUS is designed around the insight that these conditions share upstream drivers. By targeting network dysregulation rather than individual symptoms, we enable interventions with impact across multiple conditions.

Validation Strategy

NEXUS will enroll approximately 300 participants across three cohorts:

- Deep-phenotyping cohort (n=50): Intensive monitoring with full sensor platform plus quarterly multi-omic sampling (microbiome sequencing, metabolic panels, bone turnover markers, inflammatory markers). Trains and validates the causal models.
- Validation cohort (n=100-150): Confirms models generalize to broader populations. Continuous sensing plus quarterly lab validation.
- Pilot efficacy trial (n=100-150): Head-to-head comparison of network-guided versus standard treatment in PCOS, with metabolic, hormonal, and bone marker endpoints.

This staged approach concentrates expensive deep phenotyping on 50 patients while generating efficacy data. Larger clinical trials would be funded by pharma partnerships after proof-of-concept.

Risks and Mitigation

Gut sensing is unproven in this population. Breath-gut correlations have been demonstrated but not validated in PCOS patients. Mitigation: We collect stool samples as ground-truth throughout. If breath sensing underperforms, stool sampling maintains gut axis coverage.

Causal models and network patterns may not generalize beyond the training cohort. Mitigation: The validation cohort (n=100-150) explicitly tests generalization before efficacy trials.

Drug repurposing candidates may fail. Network-identified targets may not translate to clinical benefit. Mitigation: We advance multiple candidates in parallel and use the network model to identify likely responders before expensive trials.

Timeline and Investment:

NEXUS is \$60-80M program over five years, structured in four phases with milestone based Go/No-Go gates:

Phases	Milestones	Investment
Phase 0 (12-18mo)	Seedling projects that de-risk key technical elements across all TAs. ISF sensing proof of concept, preliminary network models	\$8-10M
Phase 1 (12mo)	HPA/HPO sensor validation, causal model architecture, drug candidate screening	\$12-15M
Phase 2 (12 mo)	Gut axis expansion, modeling training on cohort data, preclinical studies.	\$15-20M
Phase 3 (24mo)	Full system integration, pilot efficacy trials, IND-enabling studies, clinical validation and transition	\$25-35M

Team

Program Director: Maryam Ziaei, PhD. Stanford PhD in MEMS/sensors. Founded iSono Health (FDA-cleared AI wearable for breast health, Y Combinator backed). Director of Technology Transfer & Commercialization at ARPA-H. Deep expertise translating breakthrough science into regulated products. Senior Strategic Advisor: Dr. Renee Wegrzyn. Founding Director of ARPA-H (2022-2024). Former DARPA program manager. Unparalleled expertise in structuring ambitious health programs.

If We Succeed

NEXUS would deliver the first platform for continuous monitoring of women's neuroendocrine health, enabling earlier detection of network dysregulation during the critical perimenopausal window when intervention is most effective. Network-guided treatment could reduce the 50%+ failure rate of current one-size-fits-all protocols. More fundamentally, by targeting upstream causes rather than downstream symptoms, each successful intervention would address not just the presenting condition but reduce risk across the full cascade: metabolic dysfunction, bone loss, cardiovascular disease, and beyond.

One patient, one intervention, multiple conditions addressed. NEXUS treats root causes, not just the symptoms.

The Ask

NEXUS is seeking anchor funders who share our vision for transforming women's health through network-level understanding. We are assembling a syndicate of aligned funders to reach an initial \$10M commitment, with a target of \$60-80M over five years.

Immediate next steps:

- Technology partnerships with leading biosensing, AI, and pharma experts (Caltech, Level Zero, Dare)
- Clinical partnerships with SWAN investigators and academic medical centers with PCOS cohorts
- Phase 0 seedling projects (\$8-10M) to de-risk key technical elements

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